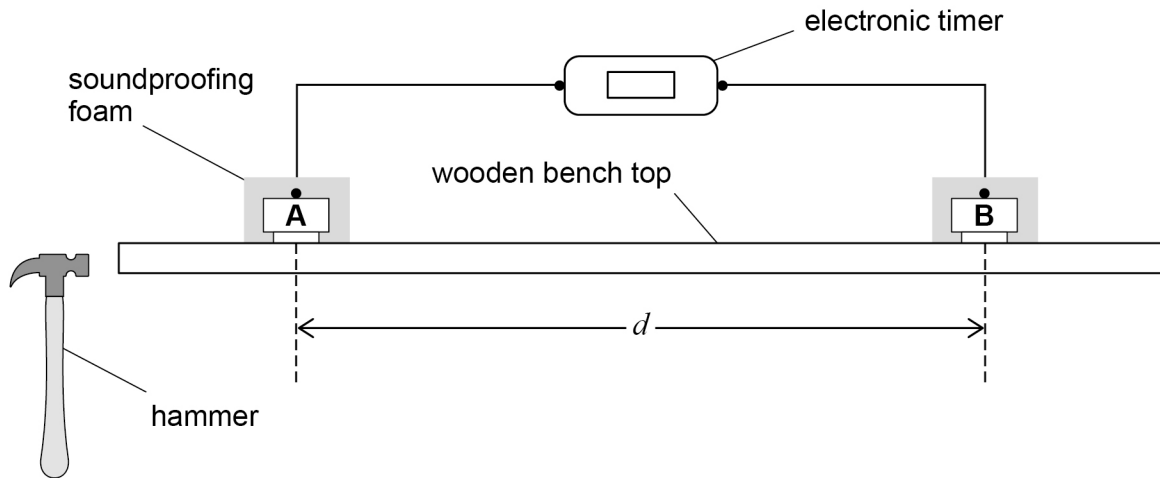


1 1

Figure 13 shows the apparatus used in an experiment to measure the speed v of sound in wood.

Figure 13



Two microphones **A** and **B**, separated by distance d , are firmly clamped face-down to the wooden bench top.

The microphones are covered with soundproofing foam and connected to an electronic timer.

The end of the bench top is struck sharply with the hammer to produce a pulse of sound which travels through the wood. The timer starts when this pulse is detected by **A**. The timer stops when the same pulse is detected by **B**.

The timer records the time t taken by the sound pulse to travel from **A** to **B**.

1 1

1

Explain why soundproofing foam is used to cover the microphones.

[1 mark]



The measurement of t is repeated several times. **Table 1** shows the recorded values of t .

Table 1

$t / \mu\text{s}$	174	196	201	179	204
-------------------	-----	-----	-----	-----	-----

1 1 . 2 Calculate the mean value of t .

[1 mark]

mean value of $t =$ _____ μs

1 1 . 3 Estimate the absolute uncertainty in your mean value of t .

[1 mark]

absolute uncertainty in $t =$ _____ μs

1 1 . 4 The distance d is 0.75 m.

Calculate v .

[1 mark]

$v =$ _____ m s^{-1}

Question 11 continues on the next page

Turn over ►



1 1 . 5 The absolute uncertainty in d is ± 0.01 m.

Estimate the percentage uncertainty in v .

[2 marks]

percentage uncertainty in v = _____

1 1 . 6 An increase in the value of d reduces the percentage uncertainty in its measurement. This increase in d also increases the value of t .

Suggest why increasing d may **not** lead to a more accurate measurement of v .

[2 marks]

8

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



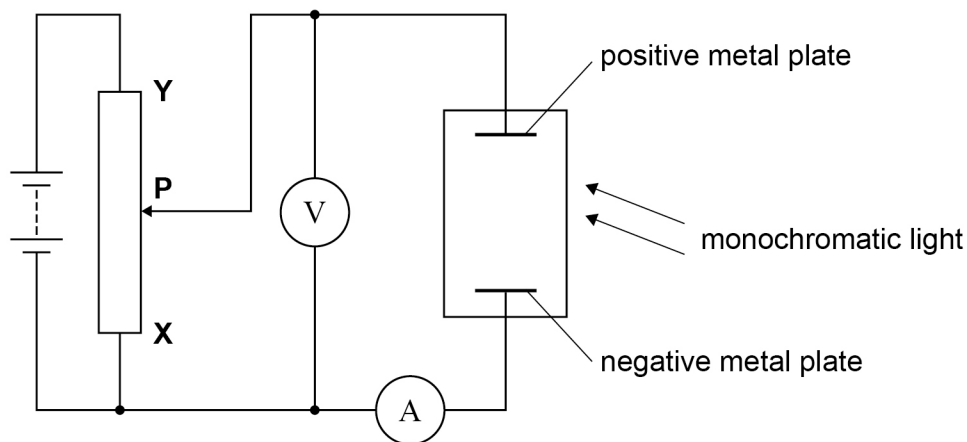
If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

A student uses the experimental arrangement shown in **Figure 13** to determine a value for the Planck constant h .

Figure 13



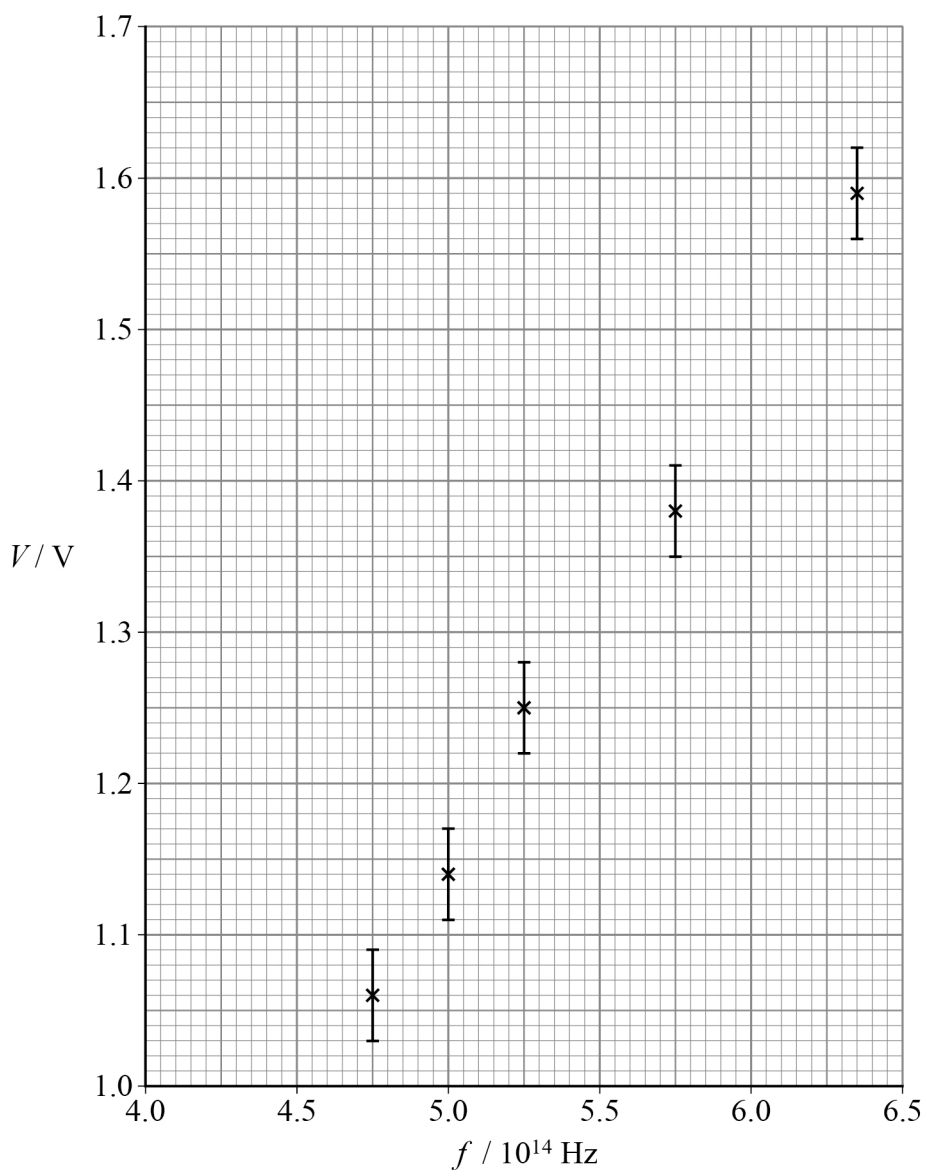
The positive metal plate is illuminated with monochromatic light of known frequency f . The ammeter detects a current.

The student then adjusts the potential difference between the plates by slowly sliding contact **P** towards **Y**. She records the minimum pd V for which the ammeter reading is zero.

The student obtains values of V for different values of f . She plots a graph of V against f as shown in **Figure 14**.



Figure 14



09.1

The student uses the absolute uncertainty in the measurements of V to plot the vertical error bars.

Determine the absolute uncertainty in the measurements of V .

[1 mark]

absolute uncertainty = _____ V

09.2

Draw, on **Figure 14**, **two** best-fit lines to show the minimum and maximum gradients consistent with the data.

[1 mark]



0 9 . 3

It can be shown that

$$eV = hf - \phi$$

Determine the minimum value of h from this experiment.

[4 marks]

minimum value of $h =$ _____ J s

0 9 . 4

Explain how the student can use her graph to work out the percentage uncertainty in her value of h .

The uncertainty in e is negligible.

[2 marks]



Don't write	Don't write
-------------	-------------

1**6**

The student measures ten oscillations of the pendulum. The stopwatch records the time for ten oscillations to be 15.07 s.

What is the best estimate for the percentage uncertainty of the time period of the pendulum?

[1 mark]

A 2.7%

☐

B 1.3%

☐

C 0.27%

☐

D 0.13%

☐

The student repeats the measurement three more times. His four measurements of the time for ten oscillations are:

15.07 s

15.35 s

15.02 s

14.92 s

What is the best estimate for the percentage uncertainty in this set of measurements?

[1 mark]

A 0.14%

☐

B 1.4%

☐

C 2.7%

☐

D 2.8%

☐

Don't waste

Figure 14 shows the seed pod of a tree. The pod consists of two seeds and two wings. When the pod falls, the wings spin to produce an upward force so that the pod travels vertically downwards at a terminal speed.

Figure 14

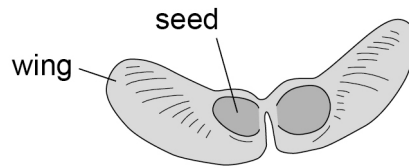
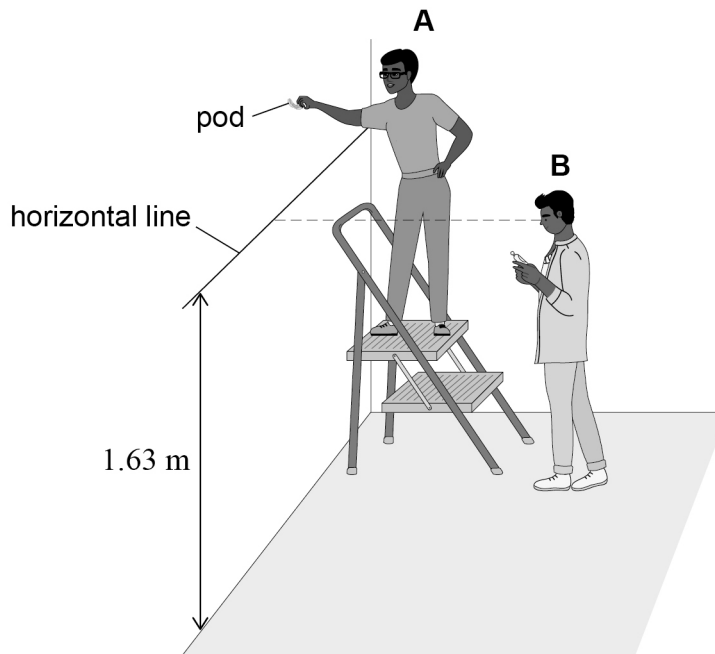


Figure 15 shows how two students **A** and **B** investigate how one pod falls.

Figure 15



Student **A** marks a horizontal line on the wall at a height of 1.63 m. The line is at the height of student **B**'s eyes when he stands facing the wall.

Student **A** releases the pod from a point higher than the line.
 Student **B** measures the time t for the pod to travel at terminal speed between the line on the wall and the floor.



The students make five valid measurements of t .

Table 1 shows four of the measurements and the mean value t_{mean} of all five measurements.

Table 1

t / s					
t_1	t_2	t_3	t_4	t_5	t_{mean}
0.84	0.75		0.94	0.77	0.83

1 0 . 1

Determine t_3 .

[1 mark]

$t_3 =$ _____ s

1 0 . 2

Calculate the percentage uncertainty in t_{mean} .

[2 marks]

percentage uncertainty = _____

1 0 . 3

Determine the absolute uncertainty in the mean terminal speed of the pod.
Assume that the uncertainty in the height is negligible.

[2 marks]

absolute uncertainty = _____ m s^{-1}

Question 10 continues on the next page

Turn over ►



1 0 . 4

The investigation is now repeated with student **B** dropping the pod and student **A** measuring the time.

Student **A** is taller than student **B**.

Suggest how the results are likely to be affected by this change.

[2 marks]

Turn over for the next question

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



Do not write

1 0 A student is asked to determine the volume of a wire that has a circular cross-section.

1 0 . 1 She makes six measurements of the diameter d of the wire.

Table 1 shows the measurements of d obtained by the student.

Table 1

d / mm	0.17	0.17	0.16	0.18	0.26	0.16
-----------------	------	------	------	------	------	------

Suggest a suitable measuring instrument to measure d .

[1 mark]

1 0 . 2 Describe how to minimise the systematic error when making these measurements.

[1 mark]

1 0 . 3 Explain why the student makes measurements of d in more than one position along the wire.

[1 mark]



1 0 . 4

The student uses **Table 1** to determine an accurate value for the cross-sectional area of the wire.

Calculate the percentage uncertainty in the cross-sectional area of the wire.

[3 marks]

percentage uncertainty = _____

1 0 . 5

The length of the wire is measured as $8.3 \text{ cm} \pm 0.1 \text{ cm}$.

Calculate the percentage uncertainty in the volume of the wire.

[2 marks]

percentage uncertainty = _____

8

Turn over for the next question

Turn over ►



Do not write